NAME: ARITRA CHAKRABORTY

DEPT.: CSE

SEMESTER: 5TH

ROLL NO.: 20

SUBJECT: DATABASE MANAGEMENT SYSTEM LAB

PAPER CODE: CS506

ASSIGNMENT: 6

PROJECT OVERVIEW:

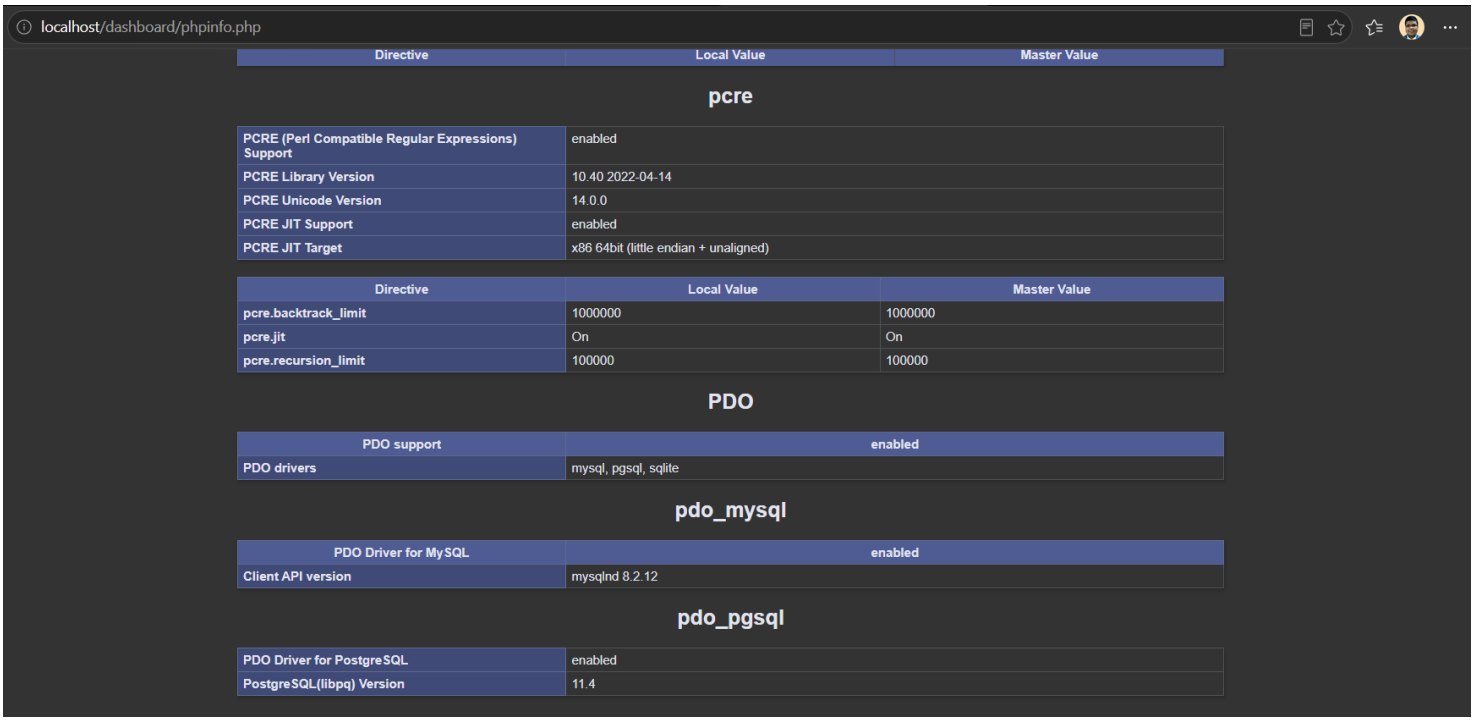
The Employee Registration Portal is a 3-tier web application designed to securely capture and persist corporate personnel details into a PostgreSQL database instance.

The frontend consists of an HTML5 form styled with external CSS providing standard input validation. The backend uses PHP with PDO (PHP Data Objects) to establish a database connection to port 5432, handle form submission securely using parameterized prepared statements to eliminate SQL injection risks, gracefully manage optional inputs with null-coalescing logic, and present user feedback upon record creation.

SUBMISSION REPORT:

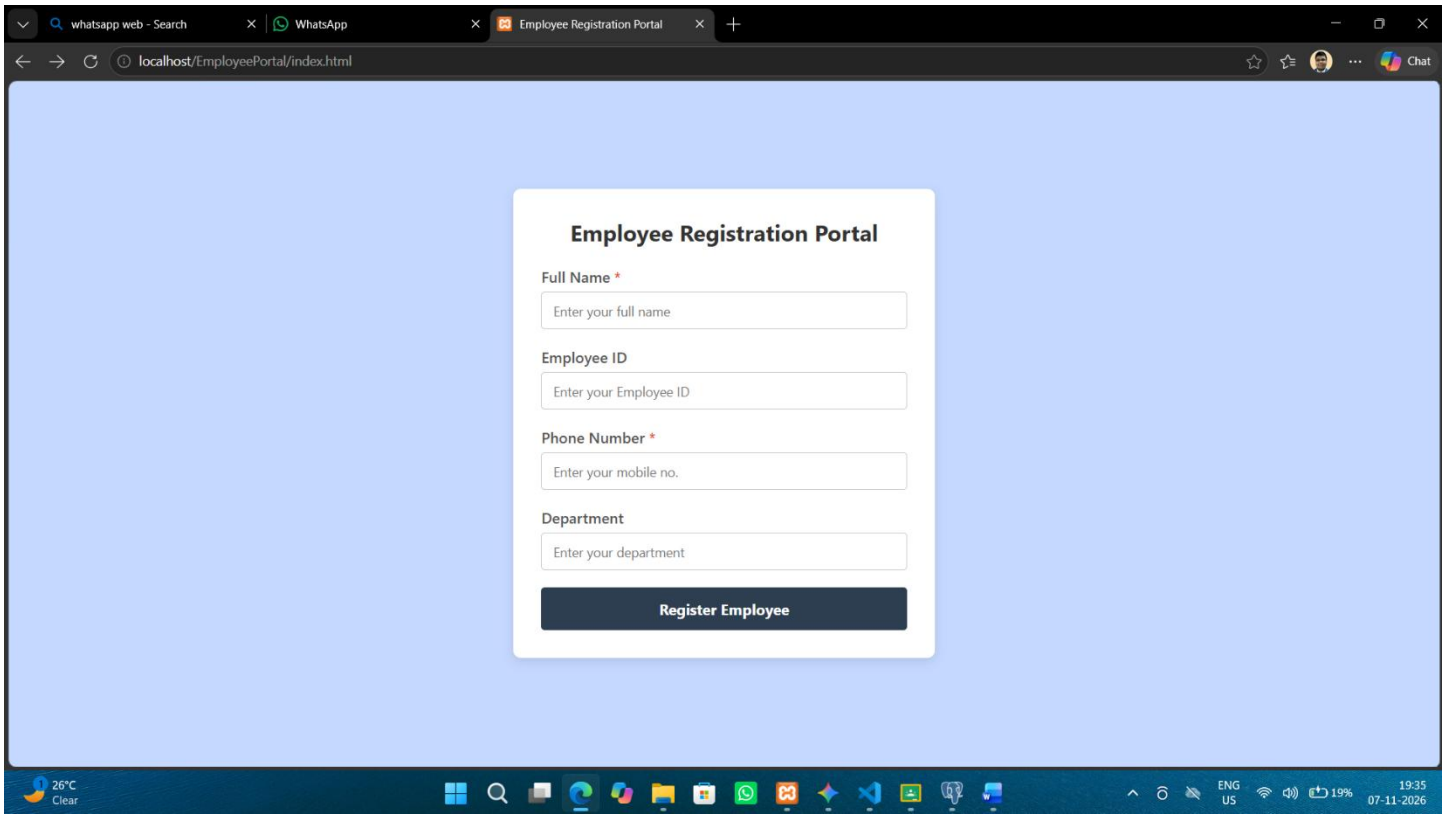
Screenshots:

1.Driver Check

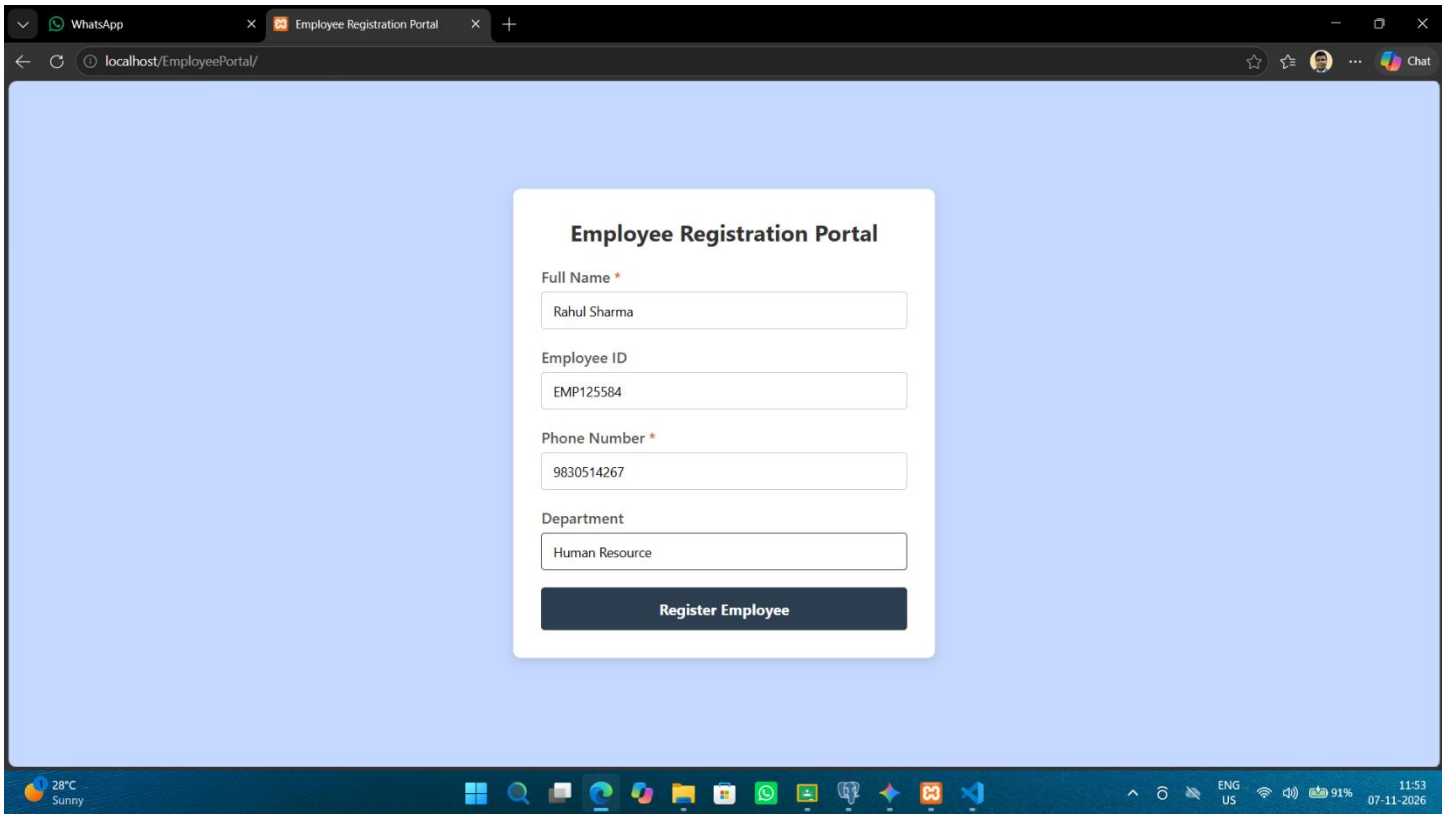


2.The form:

Blank:

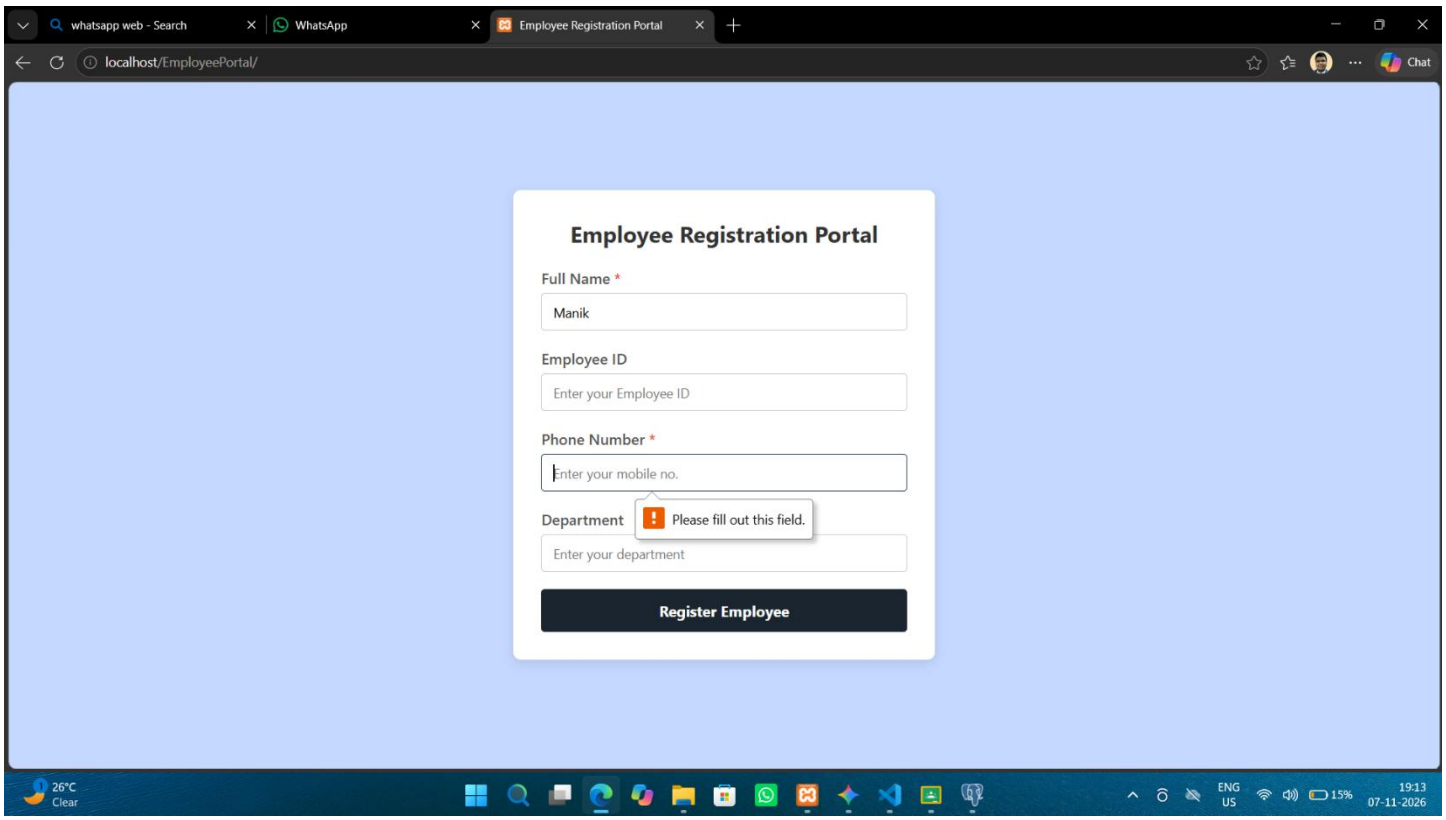


Filled Up:



3.The error:

Incomplete Submission:



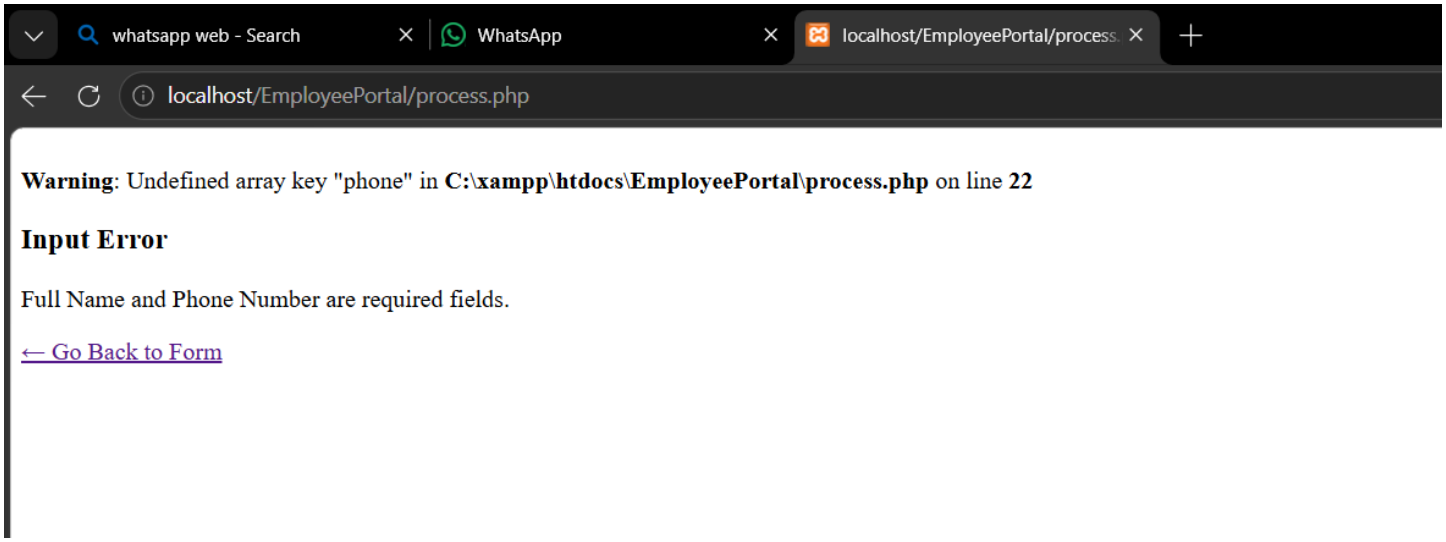
What it is: Submitting the registration form while omitting optional or mandatory field values to observe backend handling of missing input data.

Why it occurs: If the HTML form permits submission without field values and the backend directly reads $_POST array keys without check operations, uninitialized variable access throws notices or SQL null-constraint violations.

How to fix it:

- **HTML Fix:** Add the required attribute to essential inputs (emp_name and phone) so the browser prevents submission before client validation passes.
- **PHP Fix:** Apply the null coalescing operator ($_POST['field'] ?? null) or empty checks in process.php. Convert blank optional values to NULL before executing the database statement so optional columns register cleanly without errors.

Mismatched name attribute:

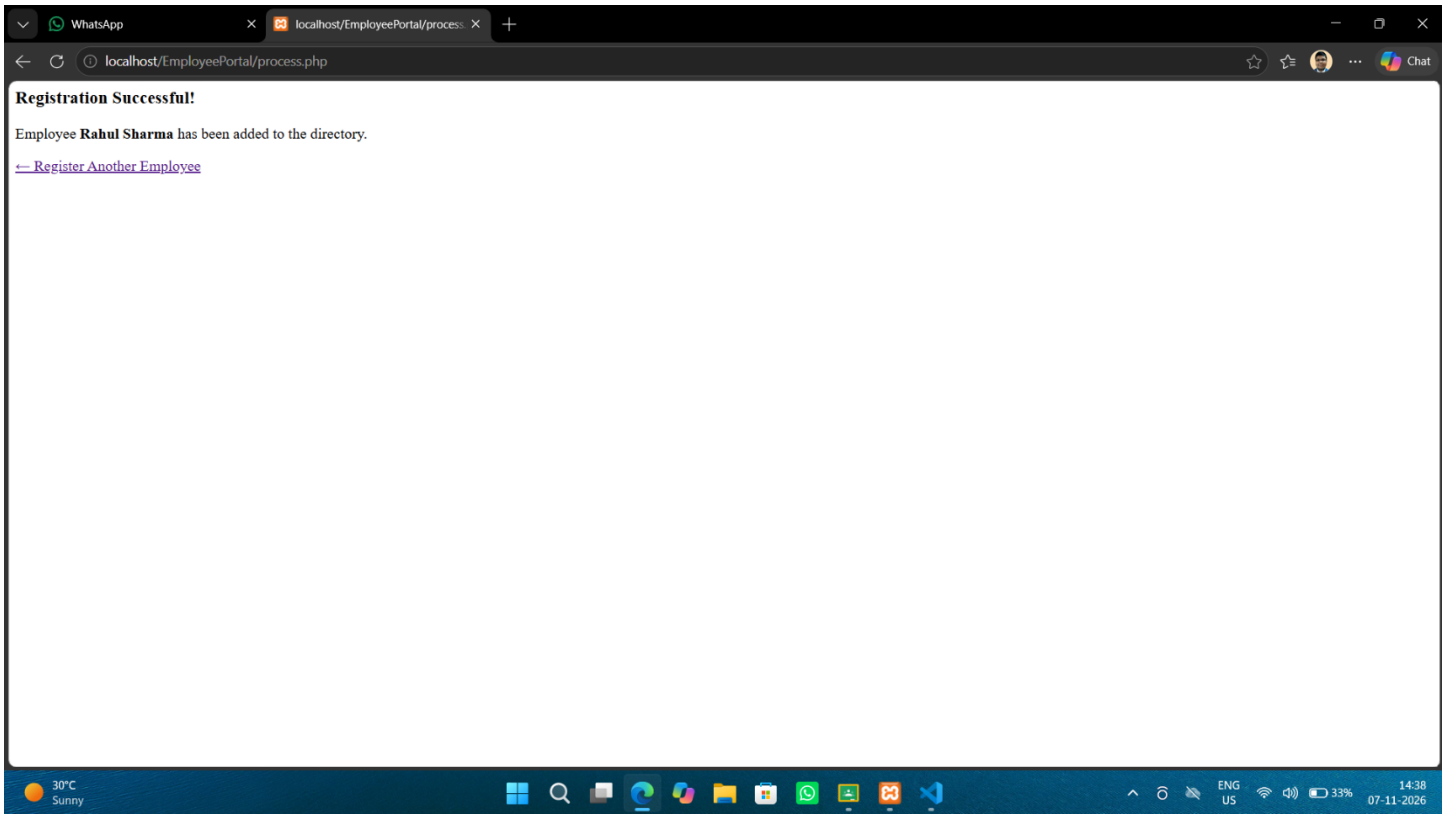


What it is: Intentionally creating a naming discrepancy between an HTML input tag's name attribute (e.g., name="Phone") and the key requested in PHP (e.g., $_POST['phone']).

Why it occurs: PHP superglobal associative arrays ($_POST) are case-sensitive. When the key names do not match identically, PHP cannot locate the submitted data key, causing a "*Warning: Undefined array key*" exception.

How to fix it: Ensure strict identifier consistency between the frontend and backend. Rename the HTML input attribute in index.html from name=" Phone" back to lowercase name="phone" to match process.php.

4.Success Response:



5. Database Verification:

Data Output | Messages | Notifications

Showing rows: 1 to 2 | Page No:

	id [PK] integer	emp_name character varying (100)	emp_id character varying (20)	phone character varying (15)	department character varying (50)	onboarded_at timestamp without time zone
1	1	Rahul Sharma	EMP125584	9830514267	Human Resource	2026-11-07 14:37:48.635669
2	2	Manik	[null]	947583112	[null]	2026-11-07 19:14:17.727293